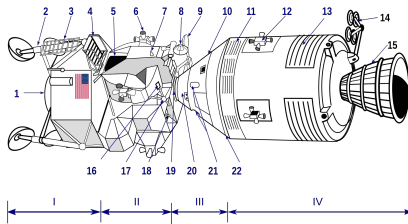


Welcome to
ASEN 3502
Computational Methods

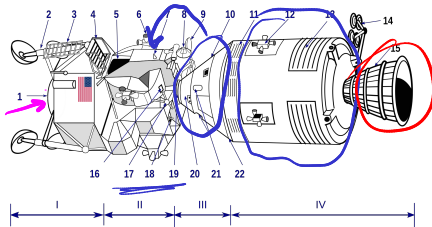
Apollo 13



I-II: lunar module (descent, ascent stages); III: command module; IV: service module.

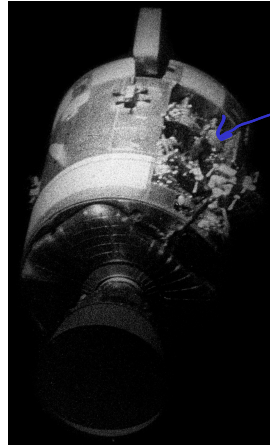
Popadius, Wikimedia Commons, CC0.

Apollo 13



I-II: lunar module (descent, ascent stages); III: command module; IV: service module.

Popadius, Wikimedia Commons, CC0.



The service module after separation, with one panel blown off. NASA (scan by Kipp Teague), public domain.

Direct return

NASA-S-66-4979 MAY 25

DIRECT RETURN MODE FOR TL COAST

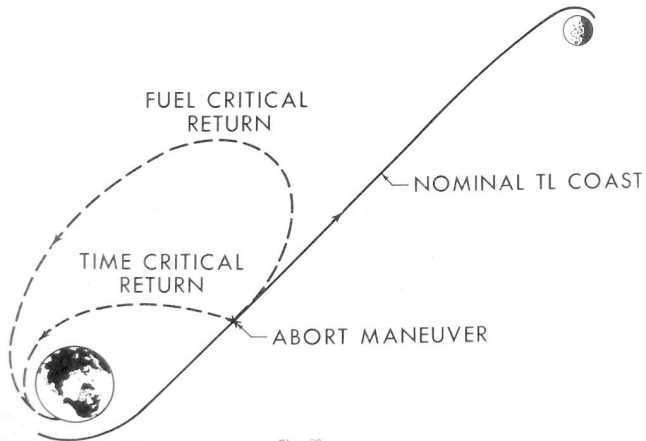
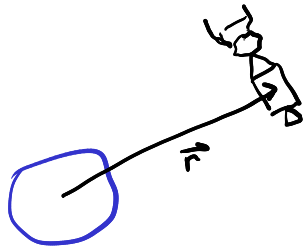


Fig. 20

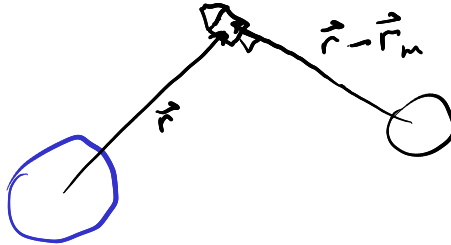
R. L. Berry, *Apollo Lunar Landing Mission Symposium*, NASA MSC, June 1966. Public domain, via Wikimedia

Commons.



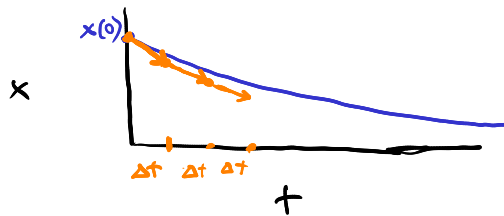
$$\ddot{\vec{r}} = -\frac{\mu_e}{r^2} \frac{\vec{r}}{r} \rightarrow \text{elliptical hyperbolic}$$

$$\mu_e = G m_e$$



$$\ddot{\vec{r}} = -\frac{\mu_e}{r^2} \frac{\vec{r}}{r} - \frac{\mu_m}{\|\vec{r} - \vec{r}_m\|^2} \frac{\vec{r} - \vec{r}_m}{\|\vec{r} - \vec{r}_m\|}$$

no easy analytical solution



$$\dot{x} = -ax$$

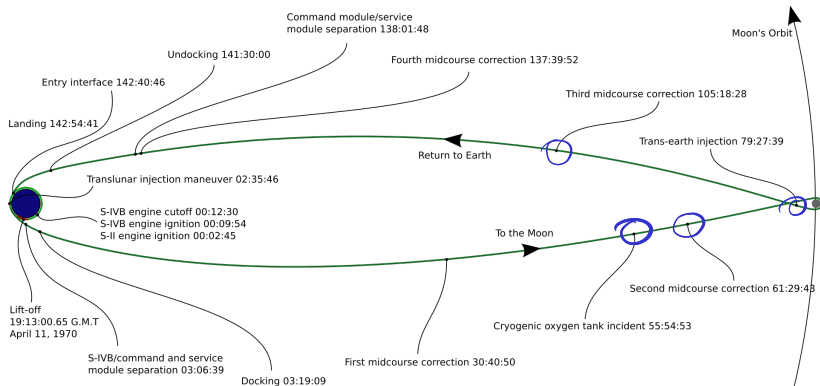
Euler's Method

$$\text{Error} = O(\Delta t)$$

Runge-Kutta

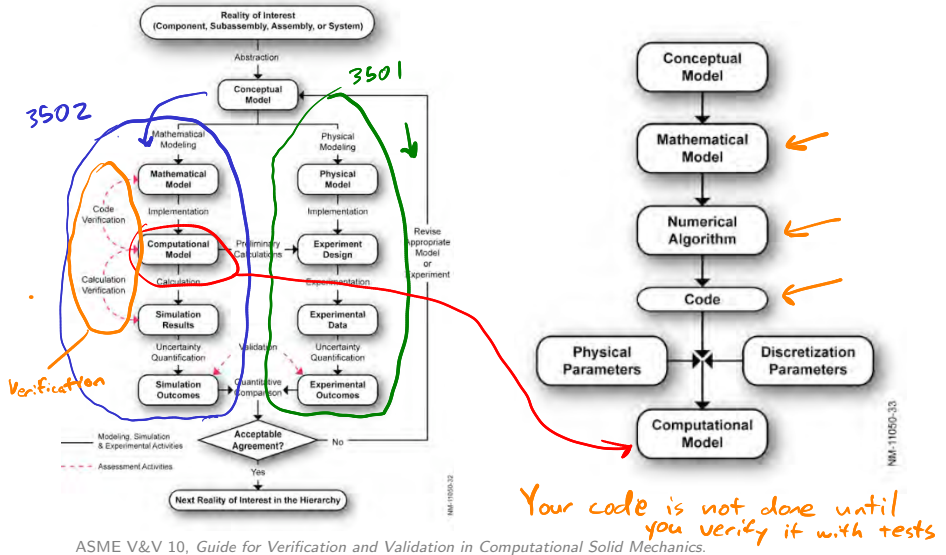
$$\text{Error} = O(\Delta t^4)$$

What Apollo 13 actually flew



AndrewBuck, Wikimedia Commons, CC BY-SA 4.0. Source for figures: Apollo 13 Mission Report, p. 3-2.

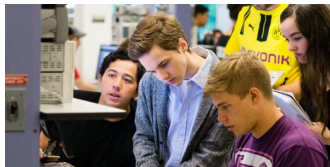
ASME V&V 10



asen3502.com

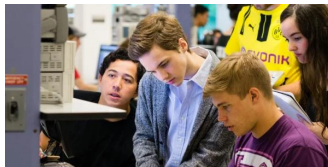
Our Assumptions

- The material is challenging, and can take multiple engagements to understand fully



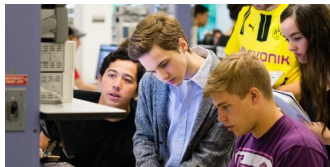
Our Assumptions

- ▶ The material is challenging, and can take multiple engagements to understand fully
- ▶ A student who comes to my office or asks a question wants to learn



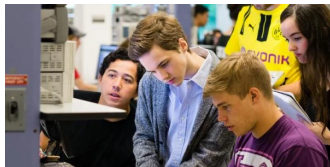
Our Assumptions

- ▶ The material is challenging, and can take multiple engagements to understand fully
- ▶ A student who comes to my office or asks a question wants to learn
- ▶ It may take multiple ways of explaining a concept before a student understands



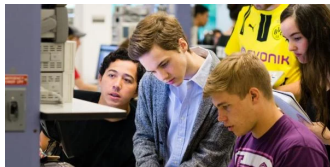
Our Assumptions

- ▶ The material is challenging, and can take multiple engagements to understand fully
- ▶ A student who comes to my office or asks a question wants to learn
- ▶ It may take multiple ways of explaining a concept before a student understands
- ▶ We must work together to help you find the way to understand



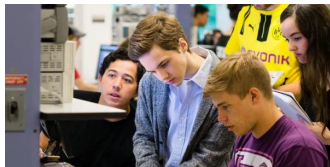
Our Assumptions

- ▶ The material is challenging, and can take multiple engagements to understand fully
- ▶ A student who comes to my office or asks a question wants to learn
- ▶ It may take multiple ways of explaining a concept before a student understands
- ▶ We must work together to help you find the way to understand
- ▶ Taking notes in class and doing the homework yourselves are critical steps to understanding



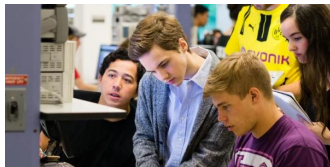
Our Assumptions

- ▶ The material is challenging, and can take multiple engagements to understand fully
- ▶ A student who comes to my office or asks a question wants to learn
- ▶ It may take multiple ways of explaining a concept before a student understands
- ▶ We must work together to help you find the way to understand
- ▶ Taking notes in class and doing the homework yourselves are critical steps to understanding
- ▶ I am trying to bring you to the point of understanding, not just give you the answer



Our Assumptions

- ▶ The material is challenging, and can take multiple engagements to understand fully
- ▶ A student who comes to my office or asks a question wants to learn
- ▶ It may take multiple ways of explaining a concept before a student understands
- ▶ We must work together to help you find the way to understand
- ▶ Taking notes in class and doing the homework yourselves are critical steps to understanding
- ▶ I am trying to bring you to the point of understanding, not just give you the answer
- ▶ We will have fun this semester



Today

- Linear Systems of Equations
- Gaussian Elim.
- Partial Pivoting

$$\begin{aligned} 2x_1 + x_2 &= 4 \\ -x_1 + x_2 &= 1 \end{aligned}$$

Unknowns: $x_1, \dots, x_n$

Linear Eqn.: $\sum_{j=1}^n a_{ij} x_j = b_i$

no nonlinear functions of x_j
 $x^2, e^x, \log(x), \sin(x)$ etc.

System of m
linear equations

$$\sum_{j=1}^n a_{ij} x_j = b_i \quad i = 1, \dots, m$$

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Book: $[A] \{x\} = \{b\}$

Handwriting: $A \vec{x} = \vec{b}$

for now $m = n$

HWO $\Rightarrow A \cdot x$

Lab

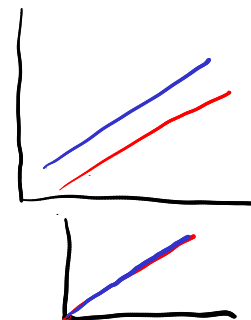
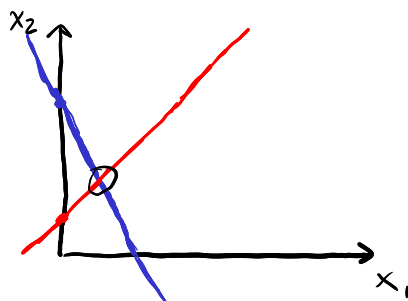
$Ax = b$
 Known $\uparrow$ Unknowns $\uparrow$ Known

$$\begin{aligned} 2x_1 + x_2 &= 4 \\ x_2 &= -2x_1 + 4 \end{aligned}$$

$$-x_1 + x_2 = 1$$

$$x_2 = x_1 + 1$$

$$x_1 = 1, x_2 = 2$$



Example:

$$\begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$3x_1 - x_2 = 0 \rightarrow 3x_1 - 3 = 0$$

$$\begin{aligned} 2x_2 &= 6 \\ x_2 &= 3 \end{aligned}$$

$$x_1 = 1$$

$$2x_1 + 3x_2 + 4x_3 = 19 \quad (1)$$

$$4x_1 + 11x_2 + 14x_3 = 55 \quad (2)$$

$$2x_1 + 8x_2 + 17x_3 = 50 \quad (3)$$

$$\begin{bmatrix} 2 & 3 & 4 & : & 19 \\ 4 & 11 & 14 & : & 55 \\ 2 & 8 & 17 & : & 50 \end{bmatrix}$$

$$\begin{aligned} & -2(1) \\ & -1(3) \end{aligned}$$

$$\begin{bmatrix} 2 & 3 & 4 & : & 19 \\ 0 & 5 & 6 & : & 17 \\ 0 & 5 & 13 & : & 31 \end{bmatrix}$$

$$-1(2)$$

$$\begin{bmatrix} 2x_1 + 3x_2 + 4x_3 = 19 \\ 0 \ 5x_2 + 6x_3 = 17 \\ 0 \ 0 \ 7x_3 = 14 \end{bmatrix}$$

$$x_3 = 2$$

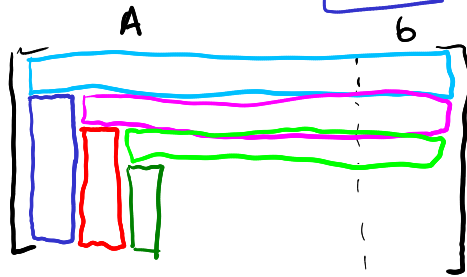
$$2x_1 + 3 \cdot 1 + 4 \cdot 2 = 19$$

$$2x_1 = 8$$

$$x_1 = 4$$

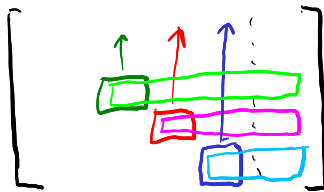
$$5x_2 + 12 = 17$$

$$x_2 = 1$$



Eliminate $\square$ with $\square$
Eliminate $\square$ with $\square$
etc

Back substitution



Solve for $\square$ with $\square$
Solve for $\square$ with $\square$
etc.

Example :

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 9 & -2 \\ 2 & 1 & -1 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 \\ 9 \\ 2 \end{bmatrix}$$

$$x = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

input : $n \times n$ matrix $[A]$, right-hand side $\{b\}$

output: solution $\{x\}$ of $[A]\{x\} = \{b\}$

// forward elimination

for $c \leftarrow 1$ to $n-1$ do

for $r \leftarrow c+1$ to n do

$$f \leftarrow A[r, c] / A[c, c]$$

$$A[r, c:n] \leftarrow A[r, c:n] - f \cdot A[c, c:n]$$

$$b[r] \leftarrow b[r] - f \cdot b[c]$$

end

end

// back substitution

for $r \leftarrow n, n-1, \dots, 1$ do

$$s \leftarrow b[r]$$

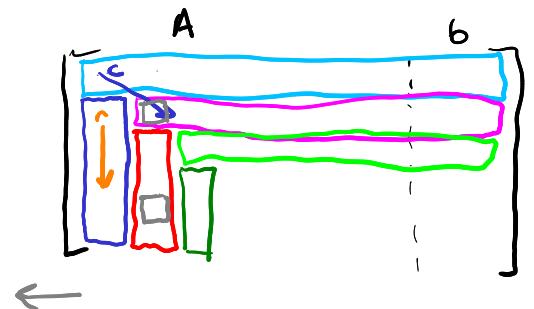
for $j \leftarrow r+1$ to n do

$$s \leftarrow s - A[r, j] \cdot x[j]$$

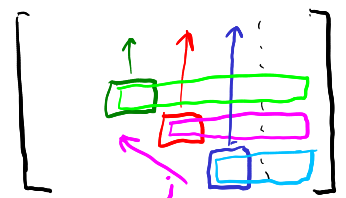
end

$$x[r] \leftarrow s / A[r, r]$$

end



Back substitution



Elimination
Backsub

$$A = \begin{bmatrix} 0 & 2 & 3 \\ 4 & 6 & 7 \\ 2 & -3 & 6 \end{bmatrix} \quad b = \begin{bmatrix} 8 \\ -3 \\ 5 \end{bmatrix}$$

$$f \leftarrow \underline{\text{Aug}[1,2]} / \underline{\text{Aug}[1,1]}$$

Divide by Zero

Solution: Partial Pivoting

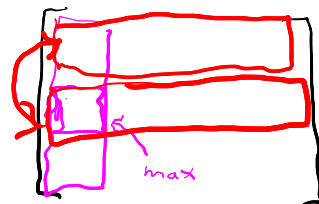
```

input :  $n \times n$  matrix  $[A]$ , right-hand side  $\{b\}$ 
output: solution  $\{x\}$  of  $[A]\{x\} = \{b\}$ 

// forward elimination
for  $c \leftarrow 1$  to  $n - 1$  do
    // find the largest entry in column  $c$ 
     $p \leftarrow \text{argmax}(\text{abs}(A[c:n, c])) + c - 1$ 
    // swap that row into the pivot position
     $A[c, :] \leftrightarrow A[p, :]$ 
     $b[c] \leftrightarrow b[p]$ 
    for  $r \leftarrow c + 1$  to  $n$  do
         $f \leftarrow A[r, c] / A[c, c]$ 
         $A[r, c:n] \leftarrow A[r, c:n] - f \cdot A[c, c:n]$ 
         $b[r] \leftarrow b[r] - f \cdot b[c]$ 
    end
end

// back substitution
for  $r \leftarrow n, n - 1, \dots, 1$  do
     $s \leftarrow b[r]$ 
    for  $j \leftarrow r + 1$  to  $n$  do
         $s \leftarrow s - A[r, j] \cdot x[j]$ 
    end
     $x[r] \leftarrow s / A[r, r]$ 
end

```



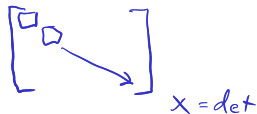
Algorithms and Operation Counting

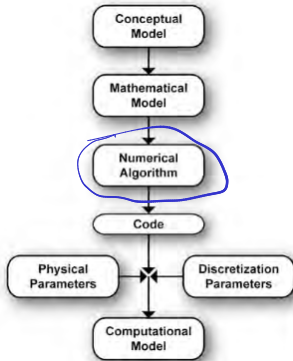
ASEN 3502 — Computational Methods

Please grab a handout in the back!

HW 1 - Due Thursday

Lab min through 1.3
 rec through 1.5





NM-11050-33

Today

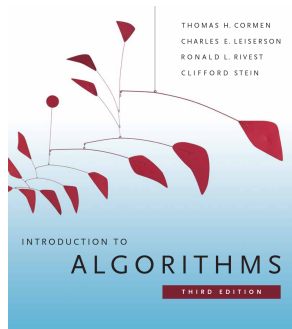
- Algorithms
- Operation Counting
- Asymptotic Notation

$O(n)$

What is an algorithm?

“Informally, an algorithm is any well-defined computational **procedure** that takes some value, or set of values, as **input** and produces some value, or set of values, as **output**.”

— Cormen et al.



Introduction to Algorithms

Cormen, Leiserson,
Rivest, and Stein

mitpress.mit.edu

FLOUR'S FAMOUS BANANA BREAD

I remember grocery shopping with my mom and taking home large bags of overripe bananas when we found them on special for ten cents a pound. My mom would encourage my brother, my dad, and me to "have a banana!" every time we were near the kitchen. My brother and I began to avoid the kitchen for fear of being scolded by Mom and her banana obsession. In time, I developed this Banana Bread as a gratification device for us. We loved it so much we sometimes found ourselves baking more bananas just to make it!

INGREDIENTS:

- 1 cup (200 grams) granulated sugar
- 2 eggs
- 1/2 cup (120 grams) vanilla oil
- 2 cups (400 grams) all-purpose flour
- 1 teaspoon baking soda
- 1/2 teaspoon ground cinnamon
- 16 medium-ripe bananas (about 2 lbs)
- 1/2 cup (100 grams) walnuts (optional)
- 1/2 cup (100 grams) raisins (optional)
- 1/2 cup (100 grams) chocolate chips (optional)

DIRECTIONS:

1. Preheat oven to 350°F (175°C).
2. Grease a loaf pan.
3. In a large bowl, mix the sugar, eggs, and vanilla oil until well combined.
4. Sift the flour, baking soda, and cinnamon into the mixture.
5. Mash the bananas in a separate bowl.
6. Fold the mashed bananas into the dry mixture.
7. If using, fold in the walnuts, raisins, or chocolate chips.
8. Pour the batter into the prepared loaf pan and smooth the top.
9. Bake for 1 to 1 1/2 hours, or until golden brown on top and the center springs back when you press it. If your finger sinks when you poke the bread, it needs to bake a little longer. Let cool in the pan on a wire rack for at least 30 minutes, then pop it out of the pan to finish cooling.
10. The banana bread can be stored tightly wrapped in plastic wrap at room temperature for up to 3 days. Or it can be well wrapped in plastic, wrapped in foil, and frozen for up to 2 weeks. Thaw overnight at room temperature for serving.

FLOUR'S FAMOUS
BANANA BREAD

Inputs

Procedure

Why algorithms?



- ▶ Algorithms are mathematical objects. Why are they useful?

Why algorithms?

- ▶ Algorithms are mathematical objects. Why are they useful?
 - ▶ We can reason about properties like correctness and resource use

Why algorithms?

- ▶ Algorithms are mathematical objects. Why are they useful?
 - ▶ We can reason about properties like correctness and resource use
 - “An algorithm is said to be **correct** if, for every input instance, it halts with the correct output.” — Cormen et al.

Why algorithms?

- ▶ Algorithms are mathematical objects. Why are they useful?
 - ▶ We can reason about properties like correctness and resource use

“An algorithm is said to be **correct** if, for every input instance, it halts with the correct output.” — Cormen et al.
 - ▶ We can make precise and rigorous statements without having to deal with issues like roundoff error, memory allocation, etc.

How do we represent algorithms?

Matrix-vector multiplication: three ways

English

To get entry r of $\{b\}$, multiply each entry in row r of $[A]$ by the matching entry of $\{x\}$, and add up the results. Do this for every row.

Code

```
b = zeros(n,1);
for r = 1:n
    for c = 1:n
        b(r) = b(r) ...
            + A(r,c)*x(c);
    end
end
```

Pseudocode

```
input      :  $n \times n$  matrix  $[A]$ ,  
              vector  $\{x\}$   
output    :  $\{b\} = [A]\{x\}$   
for  $r \leftarrow 1$  to  $n$  do  
     $b[r] \leftarrow 0$   
    for  $c \leftarrow 1$  to  $n$  do  
         $b[r] \leftarrow b[r] + A[r, c] \cdot x[c]$   
    end  
end
```

“What separates pseudocode from “real” code is that in pseudocode, we employ whatever expressive method is most clear and concise to specify a given algorithm.” — Cormen et al.

Analysis of algorithms

- ▶ *Analyzing* an algorithm means predicting the resources that an algorithm requires
 - ▶ For today, we will count floating point operations (“Flops”): multiplication, division, addition, and subtraction of floating point numbers

Analysis of algorithms

- ▶ *Analyzing* an algorithm means predicting the resources that an algorithm requires
 - ▶ For today, we will count floating point operations (“Flops”): multiplication, division, addition, and subtraction of floating point numbers
- ▶ **Discuss with your neighbor:** Consider the equation $[A]\{x\} = \{b\}$. As n gets larger, which requires more floating point operations? Finding $\{x\}$ given $[A]$ and $\{b\}$, or finding $\{b\}$ given $[A]$ and $\{x\}$?

$$\textcircled{1} \quad b = Ax$$

$$\textcircled{2} \quad Ax = b$$

Harder if using
Gaussian Elimination

Matrix-vector multiplication

input : $n \times n$ matrix $[A]$, vector $\{x\}$

output : $\{b\} = [A]\{x\}$

```
1 for r ← 1 to n do
2   b[r] ← 0
3   for c ← 1 to n do
4     b[r] ← b[r] + A[r, c] · x[c]
5   end
6 end
```

flops

0
0
2

times

$\sum_{r=1}^n 1 = n$
 $\sum_{r=1}^n \sum_{c=1}^n 1 = n^2$

$$\sum_{r=1}^n \sum_{c=1}^n 1 = \sum_{r=1}^n n = n^2$$

FLOPS: $2n^2$

Gaussian elimination

input : $n \times n$ matrix $[A]$, right-hand side $\{b\}$

output : solution $\{x\}$ of $[A]\{x\} = \{b\}$

// forward elimination

```

1 for c ← 1 to n-1 do
2   for r ← c+1 to n do
3     f ← A[r, c] / A[c, c]
4     for j ← c to n do
5       A[r, j] ← A[r, j] - f · A[c, j]
6     end
7     b[r] ← b[r] - f · b[c]
8   end
9 end

```

// back substitution

```

10 for r ← n, n-1, ..., 1 do
11   s ← b[r]
12   for j ← r+1 to n do
13     s ← s - A[r, j] · x[j]
14   end
15   x[r] ← s / A[r, r]
16 end

```

flops	times
0	$\sum_{i=1}^{n-1} 1 = n-1$
0	$\sum_{i=1}^{n-1} \frac{n^2-n}{2}$
2	$\sum_{i=1}^{n-1} \left\{ \frac{n^3}{3} + O(n^2) \right\}$
2	$\frac{n^2-n}{2}$
0	n
0	n
0	$\sum_{i=1}^{n-1} \frac{n^2-n}{2}$
2	n
1	n

Useful sum identities

$$\sum_{i=1}^m cf(i) = c \sum_{i=1}^m f(i) \quad \left| \quad \sum_{i=1}^m f(i) + g(i) = \sum_{i=1}^m f(i) + \sum_{i=1}^m g(i) \right.$$

$$\sum_{i=1}^m 1 = m$$

$$\rightarrow \sum_{i=1}^m i = \frac{m(m+1)}{2}$$

$$\sum_{i=k}^m 1 = m - k + 1$$

$$\sum_{i=k}^m i^2 = \frac{m(m+1)(2m+1)}{6} - \frac{(k-1)k(2k-1)}{6} = \frac{m^3}{3} + O(m^2)$$

$$\sum_{c=1}^{n-1} \sum_{r=c+1}^n 1 = \sum_{c=1}^{n-1} (n-c) = \sum_{c=1}^{n-1} n - \sum_{c=1}^{n-1} c$$

$$= n(n-1) - \frac{(n-1)n}{2} = \frac{n^2-n}{2}$$

$$\sum_{c=1}^{n-1} \sum_{r=c+1}^n \sum_{j=c}^n 1 = \frac{n^3}{3} - \frac{n}{3}$$

see flop-count.pdf section 2.2

<https://asen3502.com/lectures/020-algorithms/flop-count.pdf>

$$\text{FLOPS: } \frac{n^2-n}{2} + 2\left(\frac{n^3}{3} + O(n^2)\right) + 2\frac{n^2-n}{2}$$

$$\text{FLOPS: } 2\frac{n^3}{3} + O(n^2) + 2\frac{n^2-n}{2} + n$$

function $[C] = \text{matmult}([A], [B])$	flops	times
for $i = 1$ to n	0	n
for $j = 1$ to n	0	n^2
$c_{ij} = 0$	0	$\vdots$
for $k = 1$ to n	0	n^3
$c_{ij} = c_{ij} + \underline{a_{ik}} \underline{b_{kj}}$	2	n^3
end		
end	,	
end		
return $[C]$		
end		

Asymptotic notation

- ▶ Asymptotic notation is a tool for keeping track of only the terms that are most important as a function approaches an asymptote

Asymptotic notation

- ▶ Asymptotic notation is a tool for keeping track of only the terms that are most important as a function approaches an asymptote
- ▶ Chapra: $O(m^n)$ means “terms of order m^n and lower”

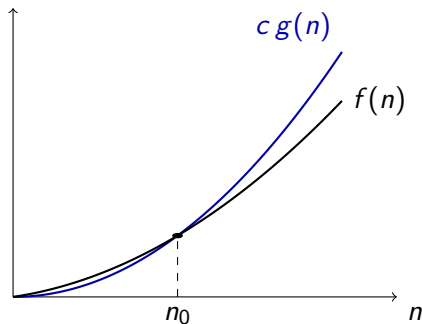
Asymptotic notation

- ▶ Asymptotic notation is a tool for keeping track of only the terms that are most important as a function approaches an asymptote
- ▶ Chapra: $O(m^n)$ means “terms of order m^n and lower”

Definition

$f(n) = O(g(n))$ as $n \rightarrow \infty$ if there exist positive constants c and n_0 such that

$$0 \leq f(n) \leq c g(n) \quad \text{for all } n \geq n_0$$



$$f(n) \leq c g(n) \quad \text{for all } n \geq n_0$$

← $f(n) = 3n^2 + 5n + 2$

$$f(n) = 3n^2 + 5n + 2$$

$$c = 4.0$$

$$n_0 = 6$$

then $0 \leq f(n) \leq c g(n)$
for all $n \geq n_0$

Therefore $f(n) = O(n^2)$

$c \geq \text{sum of coefficients}$

$$n_0 = 1$$

$$c = 10$$

$$n_0 = 1$$

Useful Rules

1) $c f(n) = O(f(n)) \quad \forall c > 0$

2) if $f_1(n) = O(n^a)$ and $f_2(n) = O(n^b)$, $f_1(n) + f_2(n) = O(n^{\max(a,b)})$

3) if $f_1(n) = O(n^a)$ and $f_2(n) = O(n^b)$, $f_1(n) \cdot f_2(n) = O(n^{a+b})$

4) if $|f(n)|$ is bounded, $f(n) = O(1)$

Example: $f(n) = \underbrace{n^2}_{O(n^2)} + \underbrace{12n}_{O(n)} \underbrace{\sin(n)}_{O(1)} + \underbrace{50}_{O(1)} = O(n^2)$



Gaussian elimination

input : $n \times n$ matrix $[A]$, right-hand side $\{b\}$

output : solution $\{x\}$ of $[A]\{x\} = \{b\}$

// forward elimination

```
for c ← 1 to n-1 do ←  $\leq n$ 
  for r ← c+1 to n do ←  $\leq n$ 
    f ←  $A[r, c] / A[c, c]$ 
    for j ← c to n do ←  $\leq n$ 
      |  $A[r, j] \leftarrow A[r, j] - f \cdot A[c, j]$ 
    end
     $b[r] \leftarrow b[r] - f \cdot b[c]$ 
  end
end
```

end

// back substitution

```
for r ← n, n-1, ..., 1 do
  s ←  $b[r]$ 
  for j ← r+1 to n do
    |  $s \leftarrow s - A[r, j] \cdot x[j]$ 
  end
   $x[r] \leftarrow s / A[r, r]$ 
end
```

end

flops	Order of times called
x	} $O(n)$
x	
✓	} $O(n^2)$
x	
✓	} $O(n^3)$
x	
✓	$O(n^2)$
x	} $O(n)$
x	
x	
✓	} $O(n^2)$
✓	$O(n)$

$O(n^3)$ Flops

Today

LU-Factorization

HW1 Due tomorrow

HW2 Assigned tonight

Lab 1: catch up with team to

1.3 (minimum)

1.5 (recommended)

before Friday

Study Hall tonight!

Recall: with a triangular matrix, T , solving $Tx=b$ for x is $O(n^2)$

Suppose we have

$LU = A$
 $\uparrow \quad \uparrow$
 lower triangular upper triangular

$$Ax = b$$

$$L(Ux) = b$$

$$\underbrace{L}_{d} \quad \underbrace{Ux}_{d}$$

0. factorize A into LU

1. solve $Ld=b$

2. solve $Ux=d$

both steps solve with a triangular matrix

$$\begin{aligned} &\Rightarrow L \quad \Rightarrow U \\ &\left(\begin{bmatrix} I \\ \begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{bmatrix} \right) \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 2 & 8 & 17 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix} \\ &\left(I + \begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \right) \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 5 & 13 \end{bmatrix} \equiv \text{II} \\ &\left(I + \begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} \right) \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix} \equiv \text{III} \\ &\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix} \\ &\quad \quad \quad L \quad \quad \quad U \end{aligned}$$

$$L = \begin{bmatrix} 1 & & & \\ f_{21} & 1 & & \\ f_{31} & f_{32} & \ddots & \\ \vdots & \vdots & \ddots & \ddots \\ f_{n1} & & & f_{nn} & 1 \end{bmatrix}$$

$U =$ result of GE

f_{ij} = factor from GE for eliminating element ij

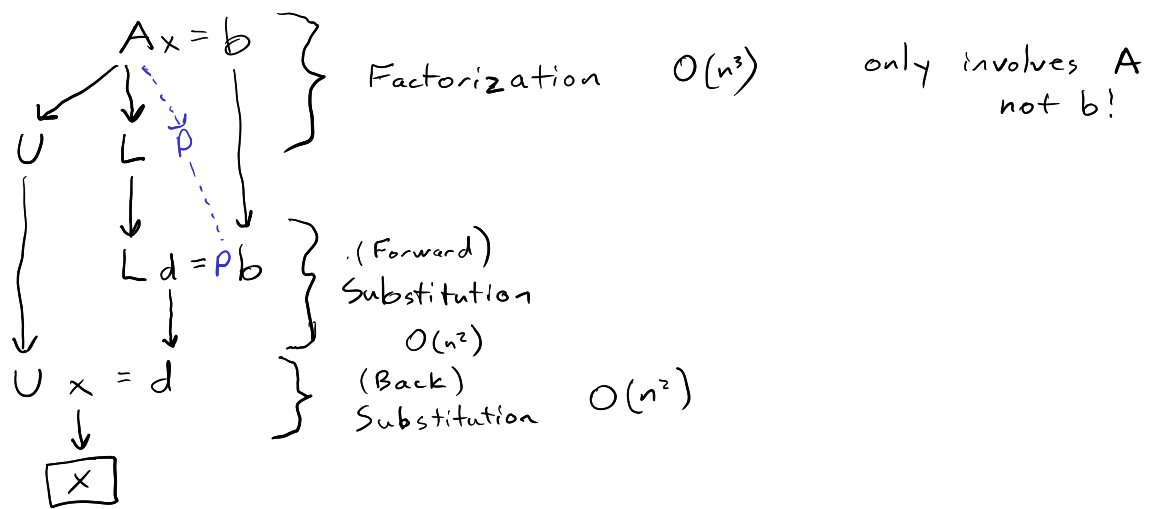
LU factorization: run GE and save factors in L

$$A = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} \begin{matrix} \textcircled{1} \\ \textcircled{2} \end{matrix} \text{ subtract } \underline{3 \times \textcircled{1}} \text{ from } 2$$

$$L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

$$LU = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} \stackrel{\checkmark}{=} A$$



If we have many b 's to solve for, factorize once ($O(n^3)$) then do forward and back substitution for each b ($O(n^2)$)!

What about Pivoting?

$$LU = PA$$

permutation matrix (keep track of pivots)

$$\begin{bmatrix} 4 & 11 & 14 \\ 2 & 3 & 4 \\ 2 & 8 & 17 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix}$$

A' P A

each row i determines which row j of A ends up in A'
 $p_{ij} = 1$ if row j of A becomes row i of A'

Exercise 2

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

$$\begin{aligned} Ax &= b \\ LU &= PA \\ \rightarrow PAx &= Pb \\ LUx &= P| \end{aligned}$$

$$\begin{bmatrix} -c- \\ -b- \\ -a- \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

Matrix Inverse Condition Number Roundoff Error

Homework 2 Due Thursday

$$\begin{aligned} Ax &= b \\ \text{inverse} \rightarrow A^{-1}b &= x \\ AA^{-1}b &= Ax = b \\ AA^{-1}b &= b \\ AA^{-1} &= I \\ &\quad (3 \times 3) \end{aligned}$$

$$A \begin{bmatrix} a_{11}^{-1} \\ a_{21}^{-1} \\ a_{31}^{-1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad A \begin{bmatrix} a_{12}^{-1} \\ a_{22}^{-1} \\ a_{32}^{-1} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad A \begin{bmatrix} a_{13}^{-1} \\ a_{23}^{-1} \\ a_{33}^{-1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Matrix Inverse via LU factorization

input : $n \times n$ matrix $[A]$

output: $[A]^{-1}$

$[L], [U] \leftarrow \text{lu_decompose}([A])$

for $k \leftarrow 1$ to n do

$\{e_k\} \leftarrow$ unit vector with 1 at element k ; 0 elsewhere
 $A^{-1}[:, k] \leftarrow \text{lu_solve}([L], [U], \{e_k\})$

end

What can go wrong?

from APPM 2360

$Ax=b$
does not
have a unique
solution

$\Leftrightarrow A$ is singular $\Leftrightarrow \det(A) = 0 \Leftrightarrow A$ is not full rank $\Leftrightarrow A^{-1}$ does not exist

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

not full rank

What if A is almost singular?

$$\text{cond}(A) = \|A\| \cdot \|A^{-1}\|$$

large cond $\Rightarrow$ nearly singular

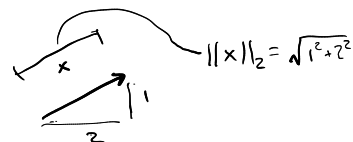
$\| \|$ means "norm"

Book covers many types of norms

For now $\| \| = \| \|_2$ "2-norm"

For a vector x , $\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$, "length" of x

For a matrix A , $\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sqrt{\lambda_{\max}}$
"The most A can elongate a vector" $\nwarrow$ max eigenvalue



Key property of cond:

$$\frac{\|\Delta x\|}{\|x\|} \leq \text{cond}(A) \frac{\|\Delta A\|}{\|A\|}$$

relative error
in result

relative error in
matrix

Roundoff Errors

Computer uses only 3 significant figures

747 takeoff mass

3.97×10^5 kg

burns 4.21 kgs fuel

$$\begin{array}{r} 397,000 \text{ kg} \\ - 4.21 \text{ kg} \\ \hline 396,995.71 \end{array} \rightarrow 3 \text{ sig figs} \rightarrow 3.97 \times 10^5 \text{ kg}$$

How computers store numbers: bits (1 or 0)

Humans think in decimal digits
0, ..., 9

Unsigned
Integers

3

137

43

1000	100	10	1
0	0	0	3
0	1	3	7
0	0	4	3

Scientific
Notation

$$\pm s \times 10^e$$

↑ significand ↗ exponent

Computers think in Binary
0, 1

1 byte

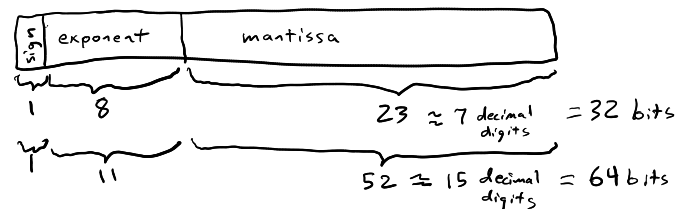
128	64	32	16	8	4	2	1
0	0	0	0	0	0	1	1
1	0	0	0	1	0	0	1
0	0	1	0	1	0	1	1

Floating Point
IEEE 754

$$\pm (1+f) \times 2^{(e-127)}$$

↑ mantissa ↗ exponent

precision
single
double



matlab/julia $\text{eps}(x)$ distance to next floating point number above x

$$\frac{\|\Delta A\|}{\|A\|} \approx \text{eps}(1.0)$$

$$\frac{\|\Delta x\|}{\|x\|} \leq \text{cond}(A) \epsilon$$

Please Grab a handout in the back.

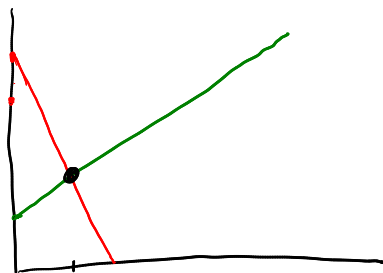
Today

Solving Nonlinear Function
Bracketing Methods: Most important: Bisection

Announcements / Reminders

Lab 1 due Thursday - Check Auto-graded!
HW 1 Released (Graded)

Linear

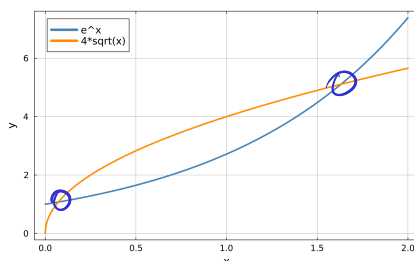


$$-2x + 4 = x + 1$$

$$x_2 = -2x_1 + 4 \quad x_2 = x_1 + 1$$

$$x = 1$$

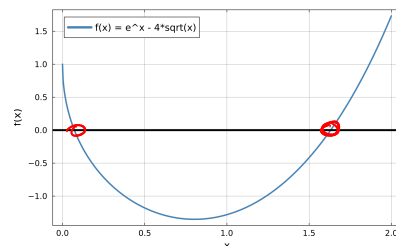
Nonlinear



$$e^x = 4\sqrt{x}$$

no analytical in "simple functions"

⇒



$$f(x) = 0 = e^x - 4\sqrt{x}$$

find "roots"
x-values where $f(x) = 0$

Bracketing

Open

↑ Today

Incremental Search

Incremental Search

input : function f , search range $[x_{\min}, x_{\max}]$, number of steps n

output: list of brackets $[x_l, x_u]$, each containing a root

$\Delta x \leftarrow (x_{\max} - x_{\min}) / n$

brackets $\leftarrow \{\}$

$x_l \leftarrow x_{\min}$

for $k \leftarrow 1$ to n do

$x_u \leftarrow x_l + \Delta x$

if $f(x_l) \cdot f(x_u) < 0$ then

// sign change $\Rightarrow$ a root lies in $[x_l, x_u]$

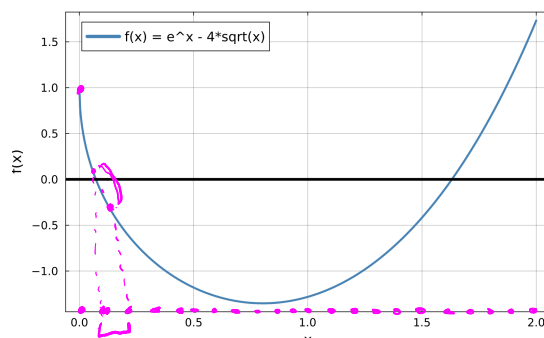
append $[x_l, x_u]$ to brackets

end

$x_l \leftarrow x_u$

end

return brackets



↑ if $f(x)$ is continuous and $f(x_l)$ and $f(x_r)$ have different signs, a root is between x_l and x_r

Strength: Can find multiple roots

Weakness: Can miss roots, slow at finding one root

(we will see shortly)

error $\rightarrow E_a = \frac{\Delta x}{n-1}$

Bisection

Bisection

input : function f , bracket $[x_l, x_u]$ with $f(x_l) \cdot f(x_u) < 0$,

tolerance ε_s

output: estimate x_r of the root

$x_r \leftarrow x_l$

repeat

$x_{r,old} \leftarrow x_r$

$x_r \leftarrow (x_l + x_u) / 2$

// keep the half that still brackets the root

if $f(x_l) \cdot f(x_r) < 0$ then

$x_u \leftarrow x_r$

else if $f(x_l) \cdot f(x_r) > 0$ then

$x_l \leftarrow x_r$

else

return x_r

end

$\varepsilon_a \leftarrow \frac{|x_r - x_{r,old}|}{x_r} \cdot 100\%$

until $\varepsilon_a < \varepsilon_s$

return x_r

// exact root

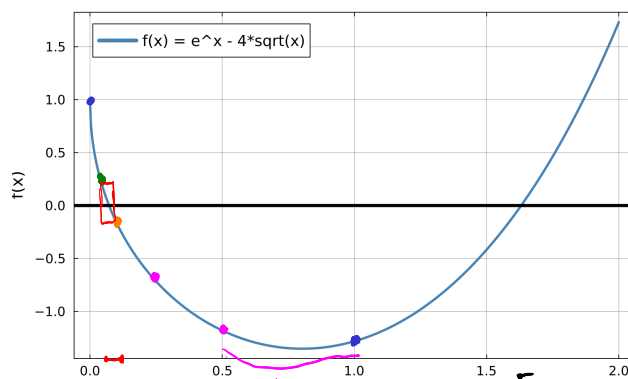
Iter: 1

2

3

4

5



same sign

E_a

$$\Delta x = 1$$

$$\Delta x = 0.5$$

$$\Delta x = 0.25$$

$$\Delta x = 0.125$$

$$\Delta x = 0.0625$$

$$\Delta x = 0.03125$$

$$E_a = \frac{\Delta x}{2^n}$$

$$E_a = \frac{\Delta x}{2^n}$$

function calls
for desired $\frac{\Delta x}{E_s}$

Incremental

$$E_a = E_s = \frac{\Delta x}{n-1}$$

$$n-1 = \frac{\Delta x}{E_s}$$

$$n = \frac{\Delta x}{E_s} + 1$$

$$O\left(\frac{\Delta x}{E_s}\right)$$

Bisection

$$E_a = E_s = \frac{\Delta x}{2^n}$$

$$2^n = \frac{\Delta x}{E_s}$$

$$n = \log_2\left(\frac{\Delta x}{E_s}\right)$$

$$O\left(\log\left(\frac{\Delta x}{E_s}\right)\right)$$

$$E_s = 0.01 \Delta x$$

$$\frac{\Delta x}{E_s} = 100$$

101 func
evals

$$\log(100) = 6.64$$

7 function
evals !

$$E_s = 0.0001 \Delta x$$

$$\frac{\Delta x}{E_s} = 10,000$$

10,001 func
evals

14 func
evals

False Position

False Position

input : function f , bracket $[x_l, x_u]$ with $f(x_l) \cdot f(x_u) < 0$,
tolerance ε_s

output: estimate x_r of the root

$x_r \leftarrow x_l$

repeat

$x_{r,old} \leftarrow x_r$

$$x_r \leftarrow x_u - \frac{f(x_u)(x_l - x_u)}{f(x_u) - f(x_l)}$$

// keep the side that still brackets the root

if $f(x_l) \cdot f(x_r) < 0$ then

| $x_u \leftarrow x_r$

else if $f(x_l) \cdot f(x_r) > 0$ then

| $x_l \leftarrow x_r$

else

| return x_r

end

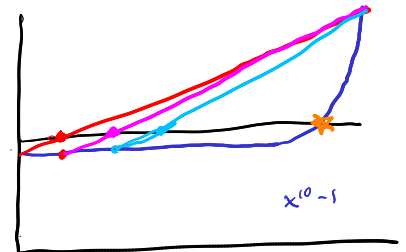
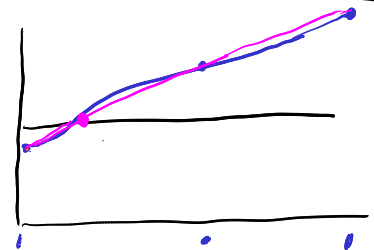
$$\varepsilon_a \leftarrow \left| \frac{x_r - x_{r,old}}{x_r} \right| \cdot 100\%$$

until $\varepsilon_a < \varepsilon_s$

return x_r

← only difference
from bisection

Problem: Doesn't
handle
curvature well



Today

Open Methods: Newton-Raphson + Friends

Secant, Brent's, Fixed Point Iter
(maybe)

Reminders

Lab 1 due tomorrow

First: Errors (Chapra's notation)

E : absolute error (units)

ϵ : relative (percent) error

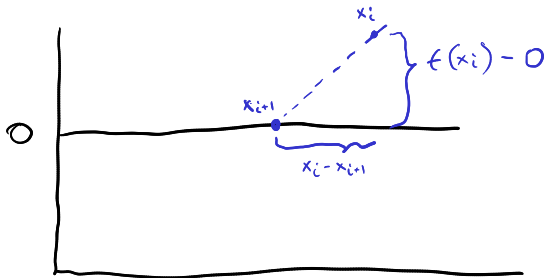
Subscripts

t : true (typically unknown)

a : approximate

s : stopping

Newton-Raphson: use $f'(x)$



$$f'(x_i) = \frac{f(x_i) - 0}{x_i - x_{i+1}} \quad \text{rise over run}$$

$$(x_i - x_{i+1}) f'(x_i) = f(x_i)$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Newton-Raphson

input: function f and its derivative f' , initial guess x_0 , tolerance ϵ_s , maximum iterations N

output: estimate x_r of the root

$x_r \leftarrow x_0$

for $i \leftarrow 1$ to N do

$x_{r,old} \leftarrow x_r$

$x_r \leftarrow x_{r,old} - \frac{f(x_{r,old})}{f'(x_{r,old})}$ ← update

$\epsilon_a \leftarrow \left| \frac{x_r - x_{r,old}}{x_r} \right| \cdot 100\%$

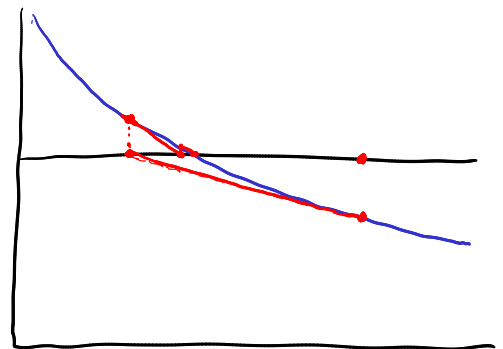
if $\epsilon_a < \epsilon_s$ then

 return x_r

end

end

return x_r // did not converge in N iterations



Beauty of Newton-Raphson: Extremely fast convergence near root!

Bisection

$$E_{a,i+1} = \frac{1}{2} E_{a,i}$$

"linear convergence"

number of sig figs increases linearly with i

(different terminology from $E_{a,i} = O((\frac{1}{2})^k)$)

Guaranteed to converge if

 - Δx contains root

 - $f(x)$ is continuous

fast

Newton Raphson

Near Root

$$\rightarrow E_{t,i+1} \approx \frac{-f''(x_r)}{2f'(x_r)} E_{t,i}^2 \quad \text{"can be shown"}$$

"quadratic convergence"

number of sigfigs doubles every iteration

Can diverge

FAST near root

Downside: Newton requires $f'(x)$

Secant Method

$$f'(x_i) \approx \frac{f(x_{i-1}) - f(x_i)}{x_{i-1} - x_i}$$

plug into Newton update

$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

Secant Method

input : function f , initial guesses x_{-1} and x_0 , tolerance ε_s , maximum iterations N

output: estimate x_r of the root

$x_{\text{prev}} \leftarrow x_{-1}; \quad x_r \leftarrow x_0$

for $i \leftarrow 1$ **to** N **do**

$x_{r,\text{old}} \leftarrow x_r$

$x_r \leftarrow x_{r,\text{old}} - \frac{f(x_{r,\text{old}})(x_{\text{prev}} - x_{r,\text{old}})}{f(x_{\text{prev}}) - f(x_{r,\text{old}})}$

$x_{\text{prev}} \leftarrow x_{r,\text{old}}$

$\varepsilon_a \leftarrow \left| \frac{x_r - x_{r,\text{old}}}{x_r} \right| \cdot 100\%$

if $\varepsilon_a < \varepsilon_s$ **then**

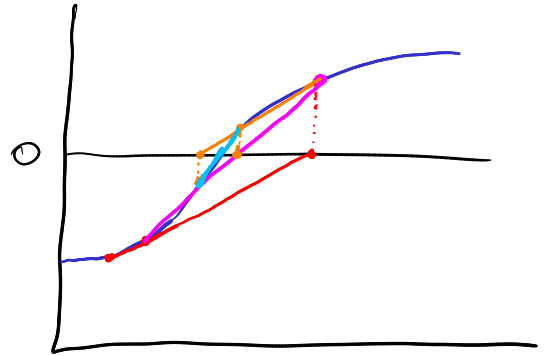
return x_r

end

end

return x_r // did not converge in N iterations

only difference from Newton



Brent's Method

Brent's Method

input : function f , bracket $[a, b]$ with $f(a) \cdot f(b) < 0$, tolerance ε_s

output: estimate b of the root

// b is the best estimate, c is on the other side of the root, a is the previous b

$c \leftarrow a$

// d is the signed step b will move by, e is the previous step;
both start as the bracket width

$d \leftarrow b - a$; $e \leftarrow d$

repeat

if $f(b) \cdot f(c) > 0$ then

 // b and c no longer bracket the root; a does

$c \leftarrow a$; $d \leftarrow b - a$; $e \leftarrow d$

end

if $|f(c)| < |f(b)|$ then

 // make b the point with the smallest $|f|$

$a \leftarrow b$; $b \leftarrow c$; $c \leftarrow a$

end

$m \leftarrow (c - b) / 2$

// bisection step

$\delta \leftarrow \varepsilon_s |b|$

// absolute tolerance

if $|m| \leq \delta$ or $f(b) = 0$ then

 return b

end

if $|e| < \delta$ or $|f(a)| \leq |f(b)|$ then

$d \leftarrow m$; $e \leftarrow m$

// bisection

else

$s \leftarrow f(b) / f(a)$

 if $a = c$ then

 // only two distinct points: secant step

$p \leftarrow 2ms$; $q \leftarrow 1 - s$

 else

 // three distinct points: inverse quadratic

 interpolation

$q \leftarrow f(a) / f(c)$; $r \leftarrow f(b) / f(c)$

$p \leftarrow s(2mq(q-r) - (b-a)(r-1))$

$q \leftarrow (q-1)(r-1)(s-1)$

 end

 if $p > 0$ then $q \leftarrow -q$

 else $p \leftarrow -p$

 if $2p < 3mq - |\delta q|$ and $p < |eq/2|$ then

 // interpolated step stays in the bracket and shrinks

 it fast enough

$e \leftarrow d$; $d \leftarrow p/q$

 else

$d \leftarrow m$; $e \leftarrow m$

// bisection

 end

end

$a \leftarrow b$

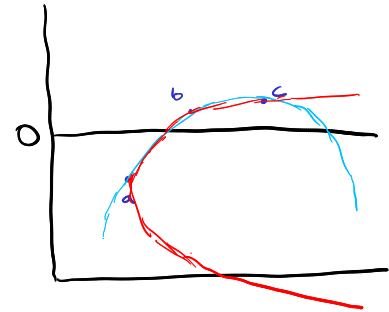
- Combination of Bracketing and Open Methods
- Does not require derivatives
- Basis for Matlab's `fzero`

- Guaranteed to converge
- As fast near root as open methods

← bisection

← secant

← inverse quadratic



Fixed Point

Previously $f(x) = 0$

Fixed Point

$$x = g(x)$$

repeatedly evaluate $g(x)$

Today

Systems of nonlinear equations

Newton-Raphson for systems

Numerical differentiation

Announcements

- HW 3 due Thurs

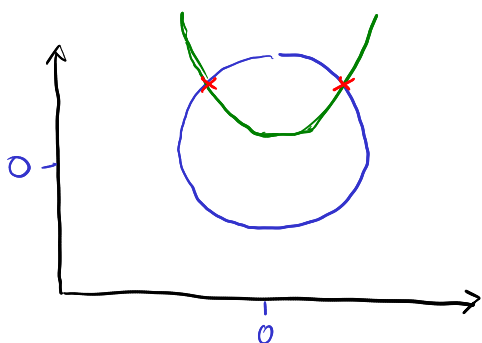
- Exam Next Monday

- Will release Eqn. Sheet/
practice exam today/tomorrow

- Calculator

- Study HW, Labs, Notes

Up to now $f(x) = 0$



$$x_1^2 + x_2^2 = 4$$

$$x_2 = x_1^2$$

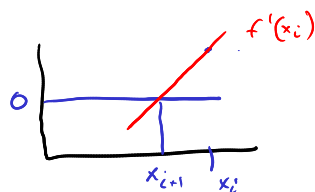
$$\{f\}(\{x\}) = \{0\}$$

$$\vec{f}(\vec{x}) = \vec{0}$$

$$\vec{f}(\vec{x}) = \begin{bmatrix} x_1^2 + x_2^2 - 4 \\ x_2 - x_1^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Newton-Raphson for scalar:

$$f(x_i) = f(x_{i+1}) + f'(x_i)(x_i - x_{i+1})$$



vector case:

first
element
of f $\rightarrow$

choose $\vec{x}_{i+1}$ so this is zero

$$f_1(\vec{x}_i) = f_1(\vec{x}_{i+1}) + \frac{\partial f_1}{\partial x_1}(x_{1,i} - x_{1,i+1}) + \frac{\partial f_1}{\partial x_2}(x_{2,i} - x_{2,i+1}) + \dots + \frac{\partial f_1}{\partial x_n}(x_{n,i} - x_{n,i+1})$$
$$= \sum_{j=1}^n \frac{\partial f_1}{\partial x_j}(x_{j,i} - x_{j,i+1})$$

$$f_2(\vec{x}_i) = \sum_{j=1}^n \frac{\partial f_2}{\partial x_j}(x_{j,i} - x_{j,i+1})$$

$\vdots$

$$f_n(\vec{x}_i) = \sum_{j=1}^n \frac{\partial f_n}{\partial x_j}(x_{j,i} - x_{j,i+1})$$

$$\vec{f}(\vec{x}_i) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \left(\begin{bmatrix} x_{1,i} \\ \vdots \\ x_{n,i} \end{bmatrix} - \begin{bmatrix} x_{1,i+1} \\ \vdots \\ x_{n,i+1} \end{bmatrix} \right)$$

$$\underbrace{\vec{f}(\vec{x}_i)}_{\text{known}} = \underbrace{J}_{\text{known}} (\underbrace{\vec{x}_i}_{\text{known}} - \underbrace{\vec{x}_{i+1}}_{\text{unknown}})$$

$$J^{-1} \vec{f}(\vec{x}_i) = \vec{x}_i - \vec{x}_{i+1}$$

$$\boxed{\vec{x}_{i+1} = \vec{x}_i - J^{-1} \vec{f}(\vec{x}_i)}$$

(scalar)

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

want $\vec{x}_{i+1} = \vec{x}_i + \Delta \vec{x}$
 $\Delta \vec{x} = \vec{x}_{i+1} - \vec{x}_i$

solve $J \Delta \vec{x} = -\vec{f}(\vec{x}_i)$
 $\vec{x}_{i+1} \leftarrow \vec{x}_i + \Delta \vec{x}$

or

$\vec{x}_{i+1} \leftarrow \vec{x}_i - J \backslash \vec{f}(\vec{x}_i)$

$A \backslash b \equiv \text{solution to } Ax = b$

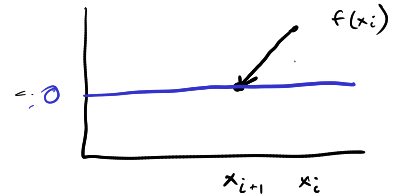
Newton-Raphson for Systems

input : function $\{f\} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and its Jacobian $[J]$ (with $J_{ij} = \partial f_i / \partial x_j$), initial guess $\{x\}_0$, tolerance ε_s , maximum iterations N

output: estimate $\{x\}_r$ of the root

```
{x}_r ← {x}_0
for i ← 1 to N do
    {x}_{r,old} ← {x}_r
    solve [J]({x}_{r,old}) {Δx} = -{f}({x}_{r,old})
    {x}_r ← {x}_{r,old} + {Δx}
    ε_a ← (||{x}_r - {x}_{r,old}|| / ||{x}_r||) · 100%
    if ε_a < ε_s then
        return {x}_r
    end
end
return {x}_r // did not converge in N iterations
```

(scalar)



Two Challenges

1. Convergence

2. Calculating Jacobian

Optimization

minimize $\sum_{k=1}^n (f_k(\vec{x}))^2$

Strength

"quadratic convergence" near root

Calculating Derivatives

Symbolic

- By hand
- Computer Algebra System (CAS)
 - e.g. Symbolic Toolbox
- Automatic Code gen

Numerical

Automatic

- Tracks exact derivative through each operation

Numerical Differentiation

Forward: $f'(x) \approx \frac{f(x+h) - f(x)}{h}$

Central: $f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$ ← more accurate

Column of Jacobian → $\frac{\partial \vec{f}}{\partial x_j} \approx \frac{\vec{f}(\vec{x} + h\vec{e}_j) - \vec{f}(\vec{x})}{h}$

size of h ?

Numerical Jacobian (forward difference)

input : function $\{f\} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, point $\{x\}$, step size h

output: approximation $[J]$ of the Jacobian at $\{x\}$, $J_{ij} \approx \partial f_i / \partial x_j$

$\{f\}_0 \leftarrow \{f(\{x\})\}$

for $j \leftarrow 1$ to n do

$\{e_j\} \leftarrow$ unit vector in the j direction // j th column of $[I]$

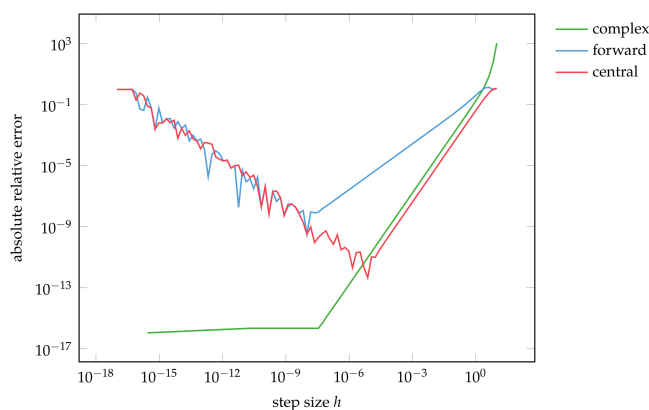
$[J][:, j] \leftarrow \frac{\{f(\{x\} + h\{e_j\})\} - \{f\}_0}{h}$ // j th column of $[J]$

end

return $[J]$

$$h \approx \sqrt{\epsilon} \times$$

↑
machine precision ϵ



Taylor Series

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)h^2}{2} + \dots$$

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \underbrace{\frac{f''(x)h}{2} + \dots}_{\text{Truncation Error}}$$

Truncation Error

Roundoff Error

$$\frac{2\epsilon |f(x)|}{h}$$

Best point is where these are equal

$$\frac{2\epsilon |f(x)|}{h} = \frac{f''(x)h}{2}$$

$$\frac{4|f(x)|}{f''(x)} \epsilon = h^2$$

$$h \approx \sqrt{\epsilon} \times$$

(right order of magnitude)

Today

Logarithmic + Exponential Growth

Reminders

- HW 3 Due Tomorrow
- Exam 1 Monday
 - Calculator, pencil
 - Be familiar w/ Eqn sheet (I will print)

Bisection requires $\log_2\left(\frac{\Delta x}{\epsilon_s}\right)$ function calls
 usually c flops

Algorithm 1

Pick up end of strip
 for $i = 1$ to L/ϵ_s
 Fold one section (accordion style)
 end

time	times executed
t_p	1
t_f	$\lceil L/\epsilon_s \rceil$

runtime
 $t_p + t_f \lceil L/\epsilon_s \rceil$
 $t_p + t_f \frac{L}{\epsilon_s} = O(L)$

Algorithm 2

1 Pick up end of strip
 2 $l_t \leftarrow L$
 3 while $l_t > \epsilon_s$
 4 ~~Run to other end~~
 5 Pick up end and fold in half
 6 $l_t \leftarrow l_t/2$
 7 end

time	times executed
t_p	1
0	1
t_f	$\log_2\left(\frac{L}{\epsilon_s}\right)$
0	

Iter	l_t
0	L
1	$\frac{L}{2}$
2	$\frac{L}{4}$
k	$\frac{L}{2^k}$

$\epsilon_s = l_t = \frac{L}{2^k}$
 $2^k = \frac{L}{\epsilon_s}$
 $k = \log_2\left(\frac{L}{\epsilon_s}\right)$

runtime:
 $t_p + t_f \log_2\left(\frac{L}{\epsilon_s}\right)$
 $= O(\log(L))$

Algorithm 2 (Running)

1 Pick up end of strip
 2 $l_t \leftarrow L$
 3 while $l_t > \epsilon_s$
 4 Run to other end
 5 Pick up end
 6 $l_t \leftarrow l_t/2$
 7 end

time	times executed
t_p	1
0	
l_t/v	$\log\left(\frac{L}{\epsilon_s}\right)$
t_p	$\log\left(\frac{L}{\epsilon_s}\right)$

runtime:
 $t_p + \sum_{j=1}^{\log_2(L/\epsilon_s)} \left(\frac{L}{2^j} \frac{1}{v} + t_p \right)$
 $= t_p \left(1 + \log_2\left(\frac{L}{\epsilon_s}\right) \right) + \sum_{j=1}^{\log_2(L/\epsilon_s)} \frac{L}{2^j} \frac{1}{v}$
 $\leq t_p \left(1 + \log_2\left(\frac{L}{\epsilon_s}\right) \right) + \frac{L}{v} \sum_{j=1}^{\infty} 2^{-j}$
 $= t_p \left(1 + \log_2\left(\frac{L}{\epsilon_s}\right) \right) + \frac{2L}{v}$
 $= O(L)$

$v = 10m/s$

Alg 1 and Alg 2(R) both have $O(L)$ runtime

$\frac{t_f}{\epsilon_s} L$ $\frac{2}{v} L$

$\frac{\epsilon_s}{t_f} = \frac{1/3 m}{1 s}$
 $\frac{t_f}{\epsilon_s} = 3$

$\frac{2}{v} = 0.2$

Alg 2 (R)
 takes $\sim 15 \times$ Alg

Logarithmic Growth

$O(1)$ $O(\log(n))$ $O(n^{\epsilon})$ $O(n^{1/2})$ $O(n)$ $O(n^2)$ $O(n^3)$
 $\log(n)$ grows slower asymptotically than any polynomial n^ϵ $\epsilon > 0$ $\nearrow O(n^\epsilon)$

$$\begin{matrix} f(n) = O(n) \\ \uparrow \\ f(n) \in O(n) \end{matrix}$$

$$O(\log(n)) \subsetneq O(n^\epsilon) \quad \forall \epsilon > 0$$

↑
strict subset

For sets of functions $O(f(n))$ and $O(g(n))$: $O(f(n)) \subsetneq O(g(n))$ means

1. $O(g(n))$ contains every function in $O(f(n))$

2. $O(g(n))$ contains at least 1 function **NOT** in $O(f(n))$

How can we prove this

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0 \Rightarrow O(f(n)) \subsetneq O(g(n))$$

$$\lim_{n \rightarrow \infty} \frac{\log(n)}{n^\epsilon} \stackrel{\text{L'Hopital}}{=} \lim_{n \rightarrow \infty} \frac{n^{-1}}{\epsilon n^{\epsilon-1}} = \lim_{n \rightarrow \infty} \frac{1}{\epsilon n^\epsilon} = 0$$

$$\therefore O(\log(n)) \subsetneq O(n^\epsilon) \quad \forall \epsilon > 0$$

$$\log_a(n) = \frac{\log_b(n)}{\log_b(a)} \quad \leftarrow \text{constant}$$

$$\therefore O(\log_a(n)) = O(\log_b(n)) \quad \forall a, b > 0$$

Exponential Growth

$a^n, a > 1$ grows faster than any polynomial

$$e^n = \sum_{j=0}^{\infty} \frac{n^j}{j!} \quad a^n = \sum_{j=0}^{\infty} \frac{(\log(a))^j n^j}{j!} \geq \frac{(\log(a))^{k+1} n^{k+1}}{(k+1)!}$$

$$\lim_{n \rightarrow \infty} \frac{n^k}{a^n} \leq \lim_{n \rightarrow \infty} \frac{(k+1)!}{(\log(a))^{k+1}} \cdot \frac{n^k}{n^{k+1}} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\therefore O(n^k) \subsetneq O(a^n) \quad \forall a > 1, k > 0$$

Let $1 < a < b$

$$O(a^n) \subsetneq O(b^n) \quad \forall 1 < a < b$$

Factorial Growth

$n!$
 $= n(n-1)(n-2)\dots 1$ grows faster than any exponential

$$\lim_{n \rightarrow \infty} \frac{a^n}{n!} = \lim_{n \rightarrow \infty} \frac{\overbrace{a \cdot a \cdot a \dots a}^N \cdot \overbrace{a \cdot a \dots a}^{n-N}}{\underbrace{1 \cdot 2 \dots N}_N \cdot \underbrace{(N+1)(N+2) \dots n}_{n-N}} \leq \lim_{n \rightarrow \infty} \frac{a^N}{N!} \left(\frac{1}{2}\right)^{n-N} = 0$$

$N = 2a$

$$O(a^n) \subsetneq O(n!) \quad \forall a > 1$$

$$O(1) \subsetneq O(\log(n)) \subsetneq O(n) \subsetneq O(n^k) \subsetneq O(a^n) \subsetneq O(n!)$$

n	$\log_2 n$	n	n^3	2^n	time	$n!$	time
10	3.3	10	10^3	10^3	μs	3.6×10^6	ms
20	4.3	20	8×10^3	10^6	ms	2.4×10^{18}	77 yr
30	4.9	30	2.7×10^4	10^9	1 s	2.7×10^{32}	10^{15} yr
60	5.9	60	2×10^5	10^{18}	36 yr	8×10^{81}	10^{65} yr

Times assume one operation per flop at 10^9 flop/s.

$\det(A)$ with Laplace Expansion requires $n!$ flops

Cramer's $A^{-1} = \frac{\text{adj}(A)}{\det(A)}$ is extremely inefficient

Today

ODEs

Euler's

Announcements
Good Luck on Exams!
Pdf + Code

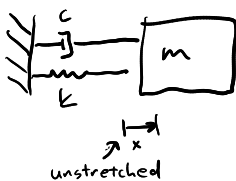
$$\dot{x} \equiv \frac{dx}{dt} = ax$$

$$\frac{dx}{dt} - ax = 0$$

$$x = C e^{at}$$

determined by boundary condition

$$x = x_0 e^{at}$$



char. eqn.

$$f = ma$$

$$-kx - c\dot{x} = m\ddot{x}$$

$$m\ddot{x} + c\dot{x} + kx = 0$$

$$m\lambda^2 + c\lambda + k = 0$$

$$\lambda = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

$$\ddot{x} = \frac{d^2x}{dt^2}$$

differential
ordinary
linear
second order
homogeneous

equation

differential: involves derivatives

ordinary: only 1 indep. variable (t) as opposed to PDE

linear: each term is linear (no x^2 , $\sin x$)

first order: only involves first derivatives

homogeneous: no "forcing" / 0 rhs

Damping

Over: λ : 2 distinct real

$$x = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$

Critically $\lambda = -\frac{c}{2m}$ (repeated)

$$x = (C_1 + C_2 t) e^{\lambda t}$$

Under $\lambda = \sigma \pm i\omega$

$$x = e^{\sigma t} (C_1 \sin(\omega t) + C_2 \cos(\omega t))$$

Roughly: solutions to linear ODEs are exponentials or growing or decaying oscillations (sin/cos)

Nonlinear ODEs ???

Numerical Integration

$$\dot{x} = f(t, x)$$

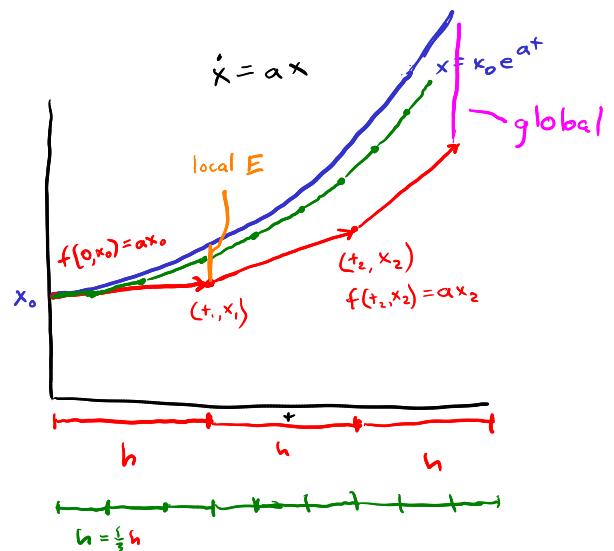
dynamics

Euler's Method

$$t_{i+1} = t_i + h$$

step size

$$x_{i+1} = x_i + f(t_i, x_i) h$$



Error Analysis

Truncation (Taylor Series)

Roundoff (floating point
sig figs)

local (each step)
global (after all steps)

Taylor Series

$$x_{i+1} = x_i + f(t_i, x_i)h + \frac{\frac{\partial f}{\partial t}(t_i, x_i)}{2!} h^2 + \frac{\frac{\partial^2 f}{\partial t^2}(t_i, x_i)}{3!} h^3 \dots$$

Euler update

$$\text{Local } E_T = \frac{\frac{\partial f}{\partial t}(t_i, x_i)}{2!} h^2 + O(h^3)$$

$$\text{Local } E_T = O(h^2) \text{ as } h \rightarrow 0$$

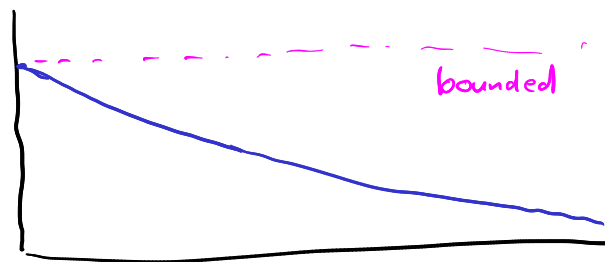
$$\text{Global } E_T = O(h) \text{ as } h \rightarrow 0$$

Stability of numerical ODE integrators

A numerical integrator is said to be unstable if errors grow exponentially for a problem with a bounded solution.

$$\dot{x} = -ax \quad a > 0$$

$$x = x_0 e^{-at}$$



Euler's $f(t, x_i)$

$$x_{i+1} = x_i - ax_i h$$
$$= x_i (1 - ah)$$

$$x_n = x_0 \underbrace{(1 - ah)^n}_{\text{amplification factor}}$$

x_n shrinks if $|1 - ah| < 1$

grows exponentially if $|1 - ah| > 1$

Stability depends on 3 factors:

- ODE
- Integration method
- step size h

For $\dot{x} = -ax \quad a > 0$

- $h > \frac{2}{a}$ Euler's unstable
- $h < \frac{2}{a}$ Euler's stable

Euler's method is conditionally stable

What about systems? higher order?

$$\rightarrow \boxed{\dot{\vec{x}} = f(t, \vec{x})}$$

$$m\ddot{x} + c\dot{x} + kx = 0$$

$$v \equiv \dot{x}$$

$$\dot{v} = -\frac{c}{m}v - \frac{k}{m}x$$

$$\vec{x} = \begin{bmatrix} x \\ v \end{bmatrix}$$

state vector

$$f(t, \vec{x}) = \begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v \\ -\frac{c}{m}v - \frac{k}{m}x \end{bmatrix}$$

state derivative

Today

Runge-Kutta

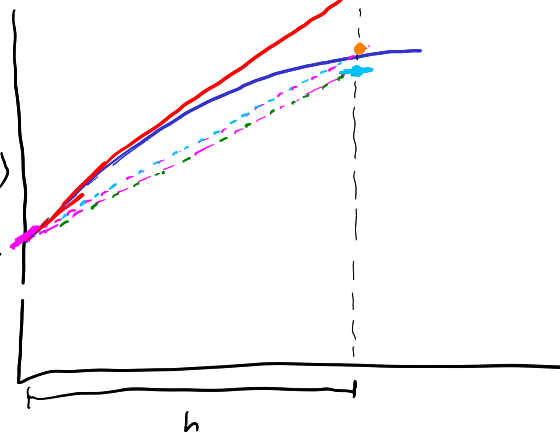
Heun's algorithm

Step i:

predict: $x_{i+1}^0 \leftarrow x_i^m + f(t_i, x_i)h$

Repeat until $\left| \frac{x_{i+1}^{j+1} - x_{i+1}^j}{x_{i+1}^j} \right| < \epsilon_s$: (n iterations over j)

correct: $x_{i+1}^j \leftarrow x_i^m + h \frac{f(t_i, x_i^m) + f(t_{i+1}, x_{i+1}^{j-1})}{2}$
 $j \leftarrow j+1$



Local: $O(h^3)$ Global: $O(h^2)$

Runge Kutta

Classical RK4 "4th order Runge Kutta"

update

$x_{i+1} \leftarrow x_i + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)h$

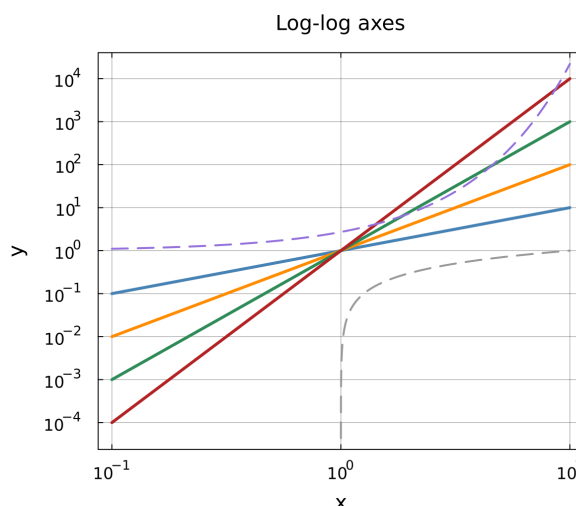
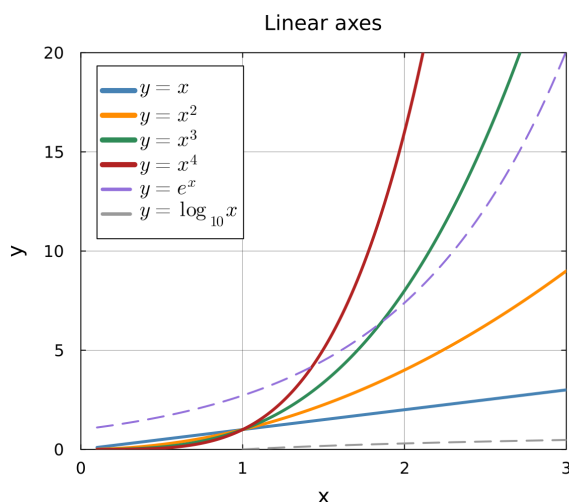
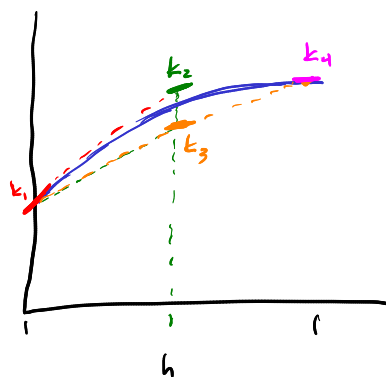
$k_1 = f(t_i, x_i)$

$k_2 = f(t_i + \frac{1}{2}h, x_i + \frac{1}{2}k_1h)$

$k_3 = f(t_i + \frac{1}{2}h, x_i + \frac{1}{2}k_2h)$

$k_4 = f(t_i + h, x_i + k_3h)$

Local: $O(h^5)$ Global: $O(h^4)$



coordinates on log-log

$x' = \log(x)$ $y' = \log(y)$

$y' = \log(x^k) = k \log(x) = kx'$

$y' = kx'$

$y = x^k$

General Runge-Kutta

$$x_{i+1} \leftarrow x_i + \phi h$$

$$\phi = a_1 k_1 + a_2 k_2 + \dots + a_n k_n$$

$$k_1 = f(t_i, x_i)$$

$$k_2 = f(t_i + p_1 h, x_i + q_{11} k_1 h)$$

$$k_3 = f(t_i + p_2 h, x_i + q_{21} k_1 h + q_{22} k_2 h)$$

$\vdots$

$$k_n = f(t_i + p_{n-1} h, x_i + q_{n-1,1} k_1 h + \dots + q_{n-1,n-1} k_{n-1} h)$$

How to find a 's, p 's, q 's?

Set equal to order- n Taylor Series

Example: Second Order

$$a_1 + a_2 = 1$$

$$a_2 p = \frac{1}{2}$$

$$a_2 q = \frac{1}{2}$$

3 equations
4 unknowns

under-determined

① $a_2 = \frac{1}{2} \Rightarrow a_1 = \frac{1}{2}, p = q = 1$

Heun's with no iter

② $a_2 = 1 \Rightarrow a_1 = 0, p = q = \frac{1}{2}$

Midpoint

③ $a_2 = \frac{3}{4} \Rightarrow a_1 = \frac{1}{4}, p = q = \frac{2}{3}$

Ralston's